During cell division chromatin is condensed into an organized form called **chromosome**. The portion an eukaryotic chromosome which is transcribed into mRNA contains active genes that are nottightly condensed during interphase is called **Euchromatin**. The portion of an eukaryotic chromosome that is not transcribed into mRNA which remains condensed during interphase and stains intensely is called **Heterochromatin. Nucleolus** is a small, dense, spherical structure either present singly or in multiples inside the nucleus and it's not membrane bound. Nucleoli possess genes for rRNA and tRNA.

**Functions of the nucleus**

- Controlling all cellular activities
- Storing the genetic or hereditary information.
- Coding the information from DNA for the production of enzymes and proteins.
- DNA duplication and transcription takes place in the nucleus.
- In nucleolus ribosomal biogenesis takes place.

### 6.7.1 Chromosomes

**Strasburger** (1875) first reported its present in eukaryotic cell and the term 'chromosome' was introduced by **Waldeyer** in 1888. **Bridges** (1916) first proved that chromosomes are the physical carriers of genes. It is made up of DNA and associated proteins.

**Structure of chromosome**

The chromosomes are composed of thread like strands called **chromatin** which is made up of DNA, protein and RNA. Each chromosome consists of two symmetrical structures called **chromatids**. During cell division the chromatids forms a well organized chromosomes with definite size and shape. They are identical and are called **sister chromatids**. A typical chromosome has narrow zones called **constrictions**. There are two types of constrictions, namely primary constriction and secondary constriction. The **primary constriction** is made up of **centromere** and kinetochore. Both the chromatids are united at centromere, whose number varies. The **monocentric** chromosome has one centromere and the **polycentric** chromosome has many centromeres. Centromere contains a complex system of protein fibres called **kinetochore**. Kinetochore is the region of chromosome which is attached to the spindle fibre during mitosis.

Besides primary there are few **secondary constrictions,** are present. Nucleoli develop from these secondary constrictions are called **nucleolar organizers**. Secondary constrictions contain the genes for ribosomal RNA which induce the formation of nucleoli and are called **nucleolar organizer regions** (Figure 6.23).

A **satellite** or SAT Chromosome is a short chromosomal segment or rounded

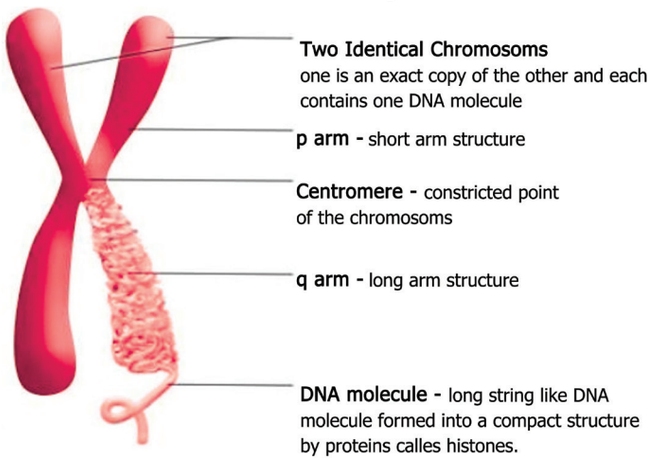



**Figure 6.23:** Structure of a Chromosome

body separated from main chromosome by a relatively elongated secondary constriction. It is a morphological entity in certain chromosomes.

**Telomere** is the terminal part of chromosome. It offers stability to the chromosome. DNA of the telomere has specific sequence of nucleotides. Telomere in all eukaryotes are composed of many repeats of short DNA sequences (**5'TTAGGG3'** sequence in *Neurospora crassa* and human beings). Maintenance of telomeres appears to be an important factor in determining the life span and reproductive capacity of



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